BIOLOGY - THE STUDY OF LIFE

Module map

- Module 01: Biology - The Study of Life
- Connected lab: lab-01_measurement-methods.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 01

Biology - The Study of Life

1. What counts as life
2. Science as a way of testing
3. Organization from cells to ecosystems
4. Evidence, variables, and measurement
5. How biological explanations improve

Lab evidence

lab-01_measurement-methods.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

BIOLOGY - THE STUDY OF LIFE

Learning objectives

- Distinguish living systems from nonliving examples using multiple characteristics.
- Separate hypotheses, predictions, observations, and conclusions in a scientific test.
- Place biological examples at the correct scale of organization.
- Identify variables and evidence in a simple investigation.
- Revise a biological explanation when new evidence appears.

OBJECTIVE 1
Distinguish living systems from nonliving examples using multiple characteristics.

OBJECTIVE 2
Separate hypotheses, predictions, observations, and conclusions in a scientific test.

OBJECTIVE 3
Place biological examples at the correct scale of organization.

OBJECTIVE 4
Identify variables and evidence in a simple investigation.

OBJECTIVE 5
Revise a biological explanation when new evidence appears.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

BIOLOGY - THE STUDY OF LIFE

Topic sequence

- What counts as life: Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.
- Science as a way of testing: Scientific claims are tested with observations, comparisons, variables, and repeatable reasoning.
- Organization from cells to ecosystems: Biological systems are organized across scales from molecules to biosphere.
- Evidence, variables, and measurement: Data become useful when measurements are tied to a clear question and source of error.
- How biological explanations improve: Strong explanations connect evidence to a claim and stay open to revision.

01

What counts as life

Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.

02

Science as a way of testing

Scientific claims are tested with observations, comparisons, variables, and repeatable reasoning.

03

Organization from cells to ecosystems

Biological systems are organized across scales from molecules to biosphere.

04

Evidence, variables, and measurement

Data become useful when measurements are tied to a clear question and source of error.

05

How biological explanations improve

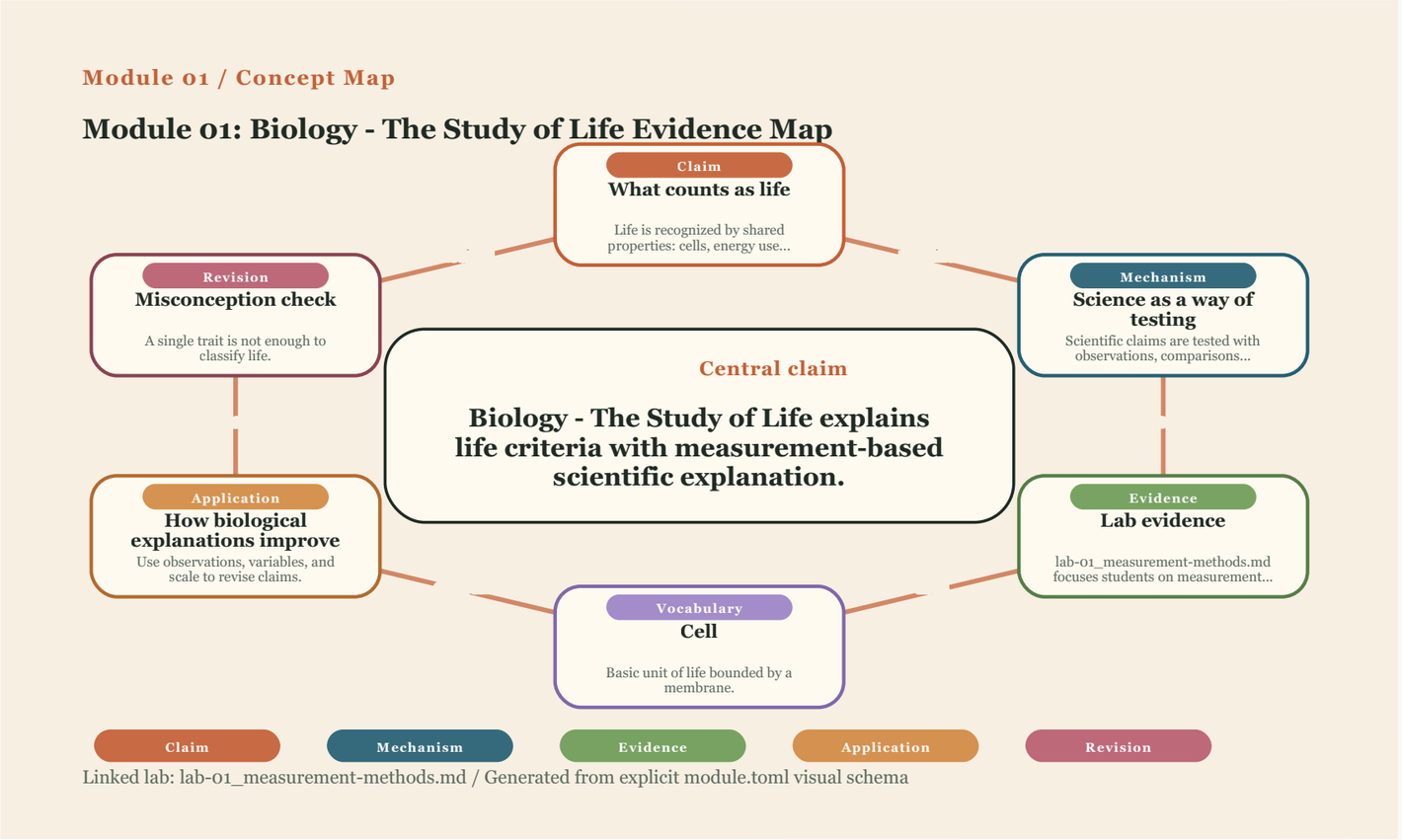
Strong explanations connect evidence to a claim and stay open to revision.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

BIOLOGY - THE STUDY OF LIFE

Module 01: Biology - The Study of Life Evidence Map

- Module focus: Biology - The Study of Life
- Central claim: Biology - The Study of Life explains life criteria with measurement-based scientific explanation.
- Key nodes to connect: What counts as life, Science as a way of testing, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-01_measurement-methods.md supplies an evidence surface for the map.

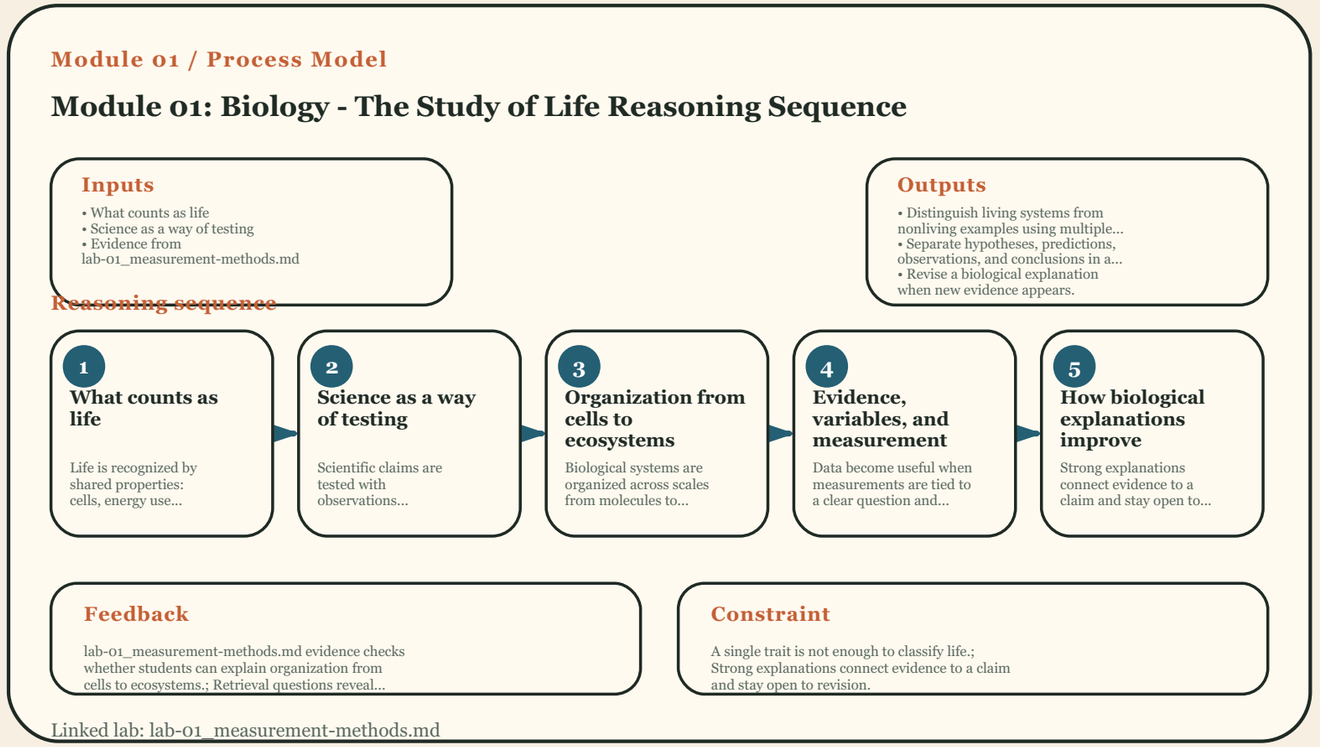


Teaching note: Ask students to explain one edge in their own words before moving on.

BIOLOGY - THE STUDY OF LIFE

Module 01: Biology - The Study of Life Reasoning Sequence

- Module focus: Biology - The Study of Life
- Trace the model: What counts as life -> Science as a way of testing -> Organization from cells to ecosystems -> Evidence, variables, and measurement
- Inputs become outputs: What counts as life, Science as a way of testing -> Distinguish living systems from nonliving examples using multiple characteristics., Separate hypotheses, predictions, observations, and conclusions in a scientific test.
- Feedback check: lab-01_measurement-methods.md evidence checks whether students can explain organization from cells to ecosystems.
- Lab connection: lab-01_measurement-methods.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

BIOLOGY - THE STUDY OF LIFE

Terms as evidence handles

- Cell: Basic unit of life bounded by a membrane.
- Homeostasis: Regulation that keeps internal conditions within a workable range.
- Metabolism: All chemical reactions that move matter and energy through life.
- Hypothesis: Testable explanation for an observed pattern.
- Variable: Factor that can change in an investigation.
- Evidence: Observation or measurement used to support or challenge a claim.

CELL
Basic unit of life bounded by a membrane.

HOMEOSTASIS
Regulation that keeps internal conditions within a workable range.

METABOLISM
All chemical reactions that move matter and energy through life.

HYPOTHESIS
Testable explanation for an observed pattern.

VARIABLE
Factor that can change in an investigation.

EVIDENCE
Observation or measurement used to support or challenge a claim.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-01_measurement-methods.md
- Lab evidence should test or illustrate: Organization from cells to ecosystems
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
What counts as life

EVIDENCE
lab-01_measurement-methods.md

CLAIM
Distinguish living systems from nonliving examples using multiple characteristics.

Teaching note: Make the lab connection explicit before students encounter the activity.

BIOLOGY - THE STUDY OF LIFE

Contrast check

- Do not stop at: Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.
- Stronger explanation: Strong explanations connect evidence to a claim and stay open to revision.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.

DEEPER

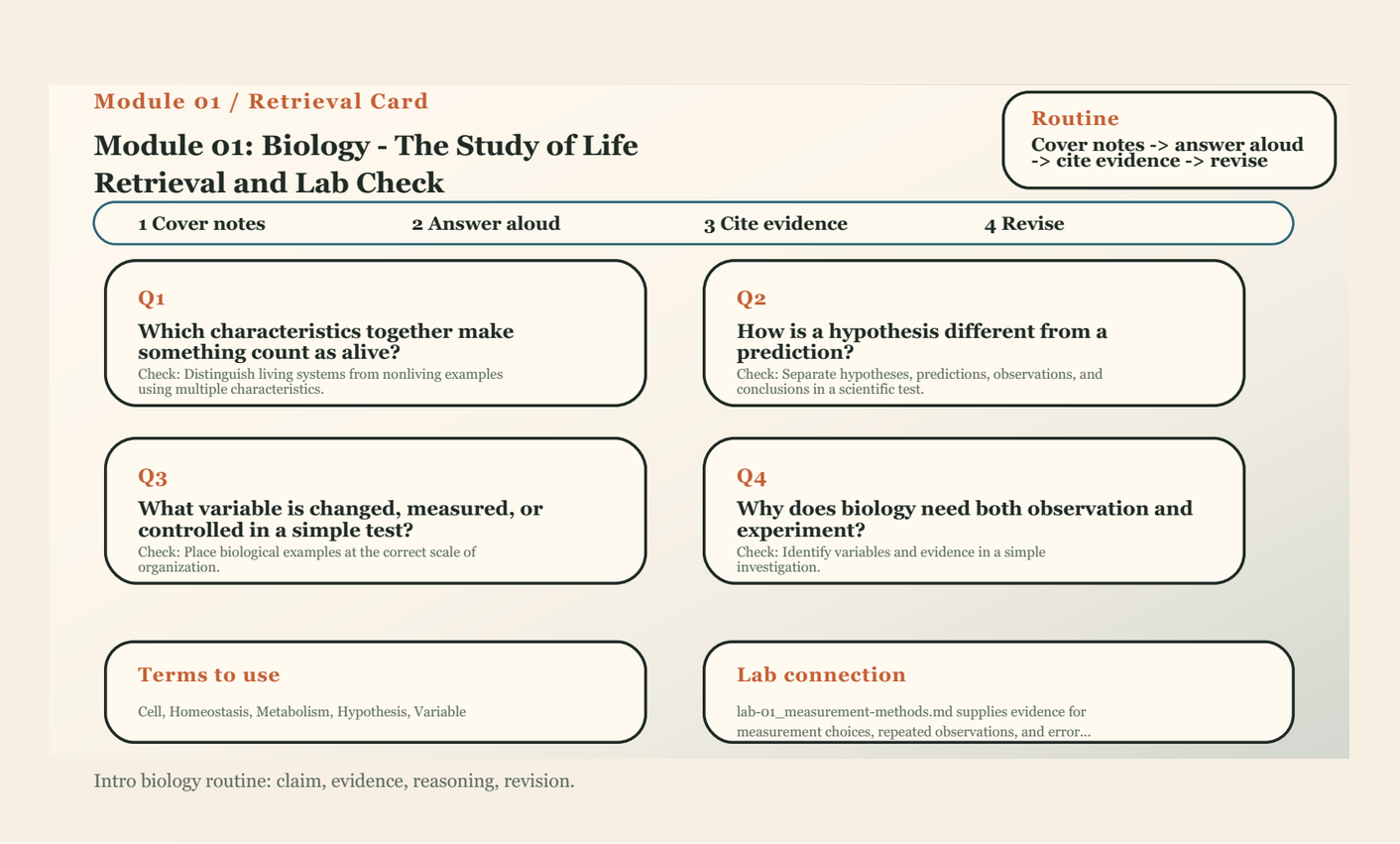
Strong explanations connect evidence to a claim and stay open to revision.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

BIOLOGY - THE STUDY OF LIFE

Module 01: Biology - The Study of Life Retrieval and Lab Check

- Module focus: Biology - The Study of Life
- Retrieval prompt: Which characteristics together make something count as alive?
- Answer check: Distinguish living systems from nonliving examples using multiple characteristics.
- Required terms: Cell, Homeostasis, Metabolism
- Lab connection: lab-01_measurement-methods.md; lab-01_measurement-methods.md supplies evidence for measurement choices, repeated observations, and error sources.



Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. A student sees frost crystals slowly "grow" on a cold window overnight. Why don't biologists count this as life?
- 2. To test whether fertilizer speeds plant growth, a student changes the amount of fertilizer given to each plant. The amount of fertilizer is the:
- 3. Which list places levels of biological organization from smallest to largest?
- 4. A hypothesis survives many careful tests over many years. What should happen to it?

QUESTION 1

Key: A

A student sees frost crystals slowly "grow" on a cold window overnight. Why don't biologists count this as life?

QUESTION 2

Key: B

To test whether fertilizer speeds plant growth, a student changes the amount of fertilizer given to each plant. The amount of fertilizer is the:

QUESTION 3

Key: C

Which list places levels of biological organization from smallest to largest?

QUESTION 4

Key: D

A hypothesis survives many careful tests over many years. What should happen to it?

Teaching note: Use the quiz as formative feedback. Answer key: 1=A, 2=B, 3=C, 4=D

BIOLOGY - THE STUDY OF LIFE

Synthesis and exit ticket

- Exit claim: explain what counts as life using evidence from lab-01_measurement-methods.md.
- Must include: Cell, Homeostasis, Metabolism.
- Revision prompt: what evidence would change your explanation?

CLAIM
What counts as life

EVIDENCE
lab-01_measurement-methods.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.